

Sakshi Patel

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SUMMARY

Software Engineer with 3+ years of experience progressing from Java-based transaction systems at LTIMindtree to Python and AI-enabled backend platforms at HCLTech. Builds secure microservices, REST APIs, event-driven workflows, and cloud-native services using Spring Boot, FastAPI, Kafka, PostgreSQL, Redis, AWS, Docker, and Kubernetes. Applies RAG, semantic search, and LLM integration to improve enterprise knowledge access while strengthening application performance, deployment reliability, automated testing, and production observability for scalable, resilient enterprise software delivery.

SKILLS

Programming Languages & Core Engineering: Python, Java, SQL, Go, Bash, Object-Oriented Programming, Data Structures, Asynchronous Programming

Backend Development & APIs: FastAPI, Spring Boot, Spring Data JPA, REST APIs, Microservices, JWT Authentication, Distributed Systems, Event-Driven Services

AI-Powered Application Development: Amazon Bedrock, LLM Integration, RAG, Vector Embeddings, Semantic Search, Amazon OpenSearch Service

Databases, Caching & Messaging: PostgreSQL, Redis, Apache Kafka, Relational Data Modeling, Database Design, Caching Strategies, Event-Driven Architecture

Cloud, Containers & CI/CD: AWS (EC2, EKS, CloudWatch), Docker, Kubernetes, GitHub Actions, Maven

Testing, Observability & Tools: Pytest, JUnit, Mockito, Unit Testing, Integration Testing, Code Coverage, Grafana, Agile/Scrum

PROFESSIONAL EXPERIENCE

Software Engineer, HCLTech

Oct 2024 - Present | Remote, USA

- Architected an AI-powered enterprise knowledge assistant using Python, FastAPI, LangChain, and OpenAI APIs, enabling employees to retrieve contextual documentation while reducing manual search effort by 34%.
- Engineered Retrieval-Augmented Generation pipelines using LangChain, Pinecone, and semantic search, enabling AI applications to deliver grounded, context-aware responses from enterprise knowledge repositories.
- Integrated secure authentication using JWT and role-based access controls, exposing REST APIs that connected React interfaces with AI services and enterprise PostgreSQL knowledge repositories.
- Optimized asynchronous inference workflows using Asyncio, request batching, and response caching, decreasing average query latency by 29% across concurrent enterprise knowledge retrieval requests.
- Automated AWS deployment pipelines for FastAPI services with GitHub Actions, validating application builds, regression testing, and infrastructure consistency before production releases across distributed environments.
- Monitored prompt performance, retrieval quality, and application behavior through structured logging, continuously refining prompts and knowledge indexing to improve successful response relevance by 22%.

Software Engineer, LTIMindtree

May 2022 - Jul 2023 | Gujarat, India

- Developed Java and Spring Boot microservices for transaction intake, validation, settlement tracking, and exception handling, exposing JWT-secured REST APIs to downstream banking applications.
- Implemented Spring Data JPA repositories and PostgreSQL schemas for transactions, merchants, settlement batches, and auditable payment records, reducing reconciliation defects by 12%.
- Integrated Apache Kafka for settlement updates, improving event-processing responsiveness by 18%, while adding validation, exception handling, and Redis caching for transaction status retrieval.
- Containerized services with Docker and supported Kubernetes deployments on AWS EC2, configuring health probes, environment properties, and CloudWatch logging under senior guidance.
- Built Maven pipelines through GitHub Actions and wrote JUnit and Mockito tests, improving release reliability by 22% across service, repository, and API layers.

PROJECTS

CI/CD Pipeline & Test Automation Framework | Python, FastAPI, Pytest, Docker, Kubernetes, GitHub Actions ([GitHub](#))

- Increased release confidence by building an automated CI/CD pipeline that runs API tests, linting, Docker builds, coverage checks, and deployment validation on every code change.
- Reduced manual deployment effort by containerizing a FastAPI service and deploying it with Kubernetes manifests, health probes, Docker smoke tests, and GitHub Container Registry publishing.

LaunchGuard - Feature Flag & Progressive Delivery Platform | Python, FastAPI ([GitHub](#))

- Reduced release risk by building a feature-flag platform that supports secure flag creation, updates, and deterministic percentage-based rollouts without requiring full application redeployments.
- Improved evaluation speed and operational visibility through PostgreSQL persistence, Redis caching, Kafka event streaming, Docker Compose infrastructure, and automated GitHub Actions testing.

EDUCATION

Master of Computer Science, Illinois Institute of Technology

Aug 2023 - May 2025 | IL, USA

Computer Science | GPA: 3.80/4.00

Bachelor of Technology, Dharmsinh Desai University

Jul 2019 - May 2023 | GJ, India

Information Technology | GPA: 3.50/4.00